Hybrid Strategy Specification

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Overview: This document specifies a hybrid trading strategy that combines two classical signals — Mean Reversion (z-score) and Cross-Sectional Momentum — under a regime-aware exposure framework. No machine learning is used; all signals are rule-based and fully interpretable.

1 Signal Definition

1.1 Signal 1 — Mean Reversion (Z-Score)

For each asset i at time t, the z-score measures how far the current price is from its recent mean, normalised by rolling standard deviation:

$$z(i,t) = \frac{\text{Close}(i,t) - \text{MA}_{20}(i,t)}{\text{StdDev}_{20}(i,t)}$$

Signal rule:

$$S_{mr}(i,t) = \begin{cases} +1, & \text{if } z(i,t) < -2 \quad \text{(oversold} \rightarrow \text{BUY)} \\ -1, & \text{if } z(i,t) > +2 \quad \text{(overbought} \rightarrow \text{SELL)} \\ 0, & \text{otherwise (HOLD)} \end{cases}$$

1.2 Signal 2 — Cross-Sectional Momentum (12-1)

For each asset i at time t, momentum is the 12-month return excluding the most recent month (to avoid short-term reversal effects):

$$\text{Mom}(i,t) = \frac{P(i,t-21)}{P(i,t-252)} - 1$$

Signal rule (cross-sectional ranking):

$$S_{mom}(i,t) = \begin{cases} +1, & \text{if rank}(\text{Mom}(i,t)) \geq \text{90th percentile (top 10\%)} \\ -1, & \text{if rank}(\text{Mom}(i,t)) \leq \text{10th percentile (bottom 10\%)} \\ 0, & \text{otherwise} \end{cases}$$

1.3 Combined Hybrid Signal

The two signals are combined with equal weighting. When both signals agree, conviction is stronger:

$$S_{hybrid}(i,t) = 0.5 \times S_{mr}(i,t) + 0.5 \times S_{mom}(i,t)$$

S_{mr}	S_{mom}	S_{hybrid}	Action
+1	+1	+1.0	Strong BUY
+1	0	+0.5	Weak BUY
0	+1	+0.5	Weak BUY
0	0	0	NEUTRAL
-1	0	-0.5	Weak SELL
0	-1	-0.5	Weak SELL
-1	-1	-1.0	Strong SELL
+1	-1	0	Conflicting → NEUTRAL

2 Prediction Horizon

2.1 Selection: 5-Day (1 Week)

We select a 5-day holding period as the primary rebalance horizon.

2.2 Justification by Signal Decay

Horizon	Signal Clarity	Cost Impact	Mean Rev	Momentum	Verdict
1D	Low	High	Poor	Poor	Rejected
5D	Medium-High	Moderate	Good	Good	Selected
10D	Medium	Low	Poor	Good	Rejected

Why 5D? Mean reversion signals typically materialise within 3–7 trading days. Momentum signals have not yet decayed. Transaction costs remain manageable at weekly rebalance vs. daily. The 5-day horizon is the sweet spot for both signals simultaneously.

3 Trading Style

3.1 Style: Cross-Sectional Ranking

At each rebalance date t, for each asset i in the universe of $N = 50$ stocks:

1. Compute $S_{hybrid}(i, t)$ for all assets.
2. Rank all assets by S_{hybrid} (descending).
3. Long basket: top 20% of assets ($\approx$ 10 stocks).
4. Short basket: bottom 20% of assets ($\approx$ 10 stocks).
5. Neutral: middle 60% → no position.

3.2 Rebalance Frequency

Weekly rebalance — every 5 trading days. Matches the 5-day holding period and mean reversion signal half-life.

Note: For a pure momentum strategy a monthly rebalance would also be valid (momentum persists over weeks), but the hybrid requires weekly frequency to capture mean reversion signals before they decay.

3.3 Position Sizing

Base position size before regime scaling (equal-weighted within each basket):

$$|w(i,t)| = \frac{1}{k} \quad \text{where } k = \text{number of selected assets per side}$$

Final adjusted weight:

$$w(i,t) = \text{signal}(i,t) \times \frac{1}{k} \times m_t$$

Note: Equal weighting $\left(\frac{1}{k}\right)$ avoids concentration risk and ensures each selected asset contributes equally to portfolio performance regardless of signal strength.

Parameter	Value	Description
Portfolio type	Long-short	Market-neutral target
Position weighting	Equal weight $(1/k)$	k = assets per side
Max position/asset	$\pm 15\%$	Of gross notional
Max gross exposure	150%	1.5× leverage
Max net exposure	$\pm 20\%$	Near market-neutral

4 Regime Filter

Two independent regime filters jointly determine the exposure multiplier m_t.

4.1 Volatility Regime

Rolling 20-day annualised volatility ($\alpha(t)$ notation from momentum strategy spec):

$$\sigma(t) = \sqrt{252} \times \text{StdDev}\left(r(t-19), \ldots, r(t)\right)$$

Regime classification uses percentile thresholds. For the pure momentum strategy (Phase 2), the 30th/70th percentiles were used. For the hybrid, we use rolling 252-day quantiles (25th/75th) to be more adaptive to changing market conditions.

Regime	Condition	$\alpha(t)$ — Momentum	Hybrid $\alpha(t)$
HIGH VOL	$\sigma >$ 70th pct (or Q_{75})	0.5	0.5
NORMAL	30th–70th pct (or Q_{25}–Q_{75})	1.0	1.0
LOW VOL	$\sigma <$ 30th pct (or Q_{25})	1.2	1.2

Rationale: Multiplier values (0.5 / 1.0 / 1.2) are identical to the momentum strategy spec. In high-volatility regimes, z-scores generate more false positives and momentum strategies suffer crashes. Reducing exposure by 50% protects capital during dislocations (e.g. March 2020, Q4 2018).

4.2 Trend Regime

Market trend defined using SPY relative to its 50-day moving average ($\beta(t)$ notation from momentum strategy spec). SPY Close at $t-1$ is used to prevent look-ahead bias:

$$\text{MA50}_{SPY}(t) = \frac{1}{50} \sum_{k=1}^{50} \text{SPY_Close}(t-k)$$

$$\text{Trend}(t) = \begin{cases} 1 & \text{(BULL)}, \quad \text{if SPY_Close}(t-1) > \text{MA50}_{SPY}(t) \\ 0 & \text{(BEAR)}, \quad \text{otherwise} \end{cases}$$

Regime	Condition	$\beta(t)$ — Momentum	Hybrid $\beta(t)$
BULL	SPY$(t-1) >$ MA50	1.0	1.0
BEAR	SPY$(t-1) \leq$ MA50	0.5	0.6

Adaptation note: The momentum strategy used $\beta(t) = 0.5$ for bear markets. For the hybrid, we use 0.6 because the mean reversion signal retains some edge in bear markets (oversold conditions still revert), so we reduce exposure less aggressively than for pure momentum.

4.3 Combined Regime Multiplier

The final position multiplier combines both regime factors (using α/β notation consistent with momentum strategy spec):

$$w(i,t) = \alpha(t) \times \beta(t) \times \widetilde{w}(i,t)$$

where

$$\widetilde{w}(i,t) = \frac{\text{signal}(i,t)}{k}$$

	BULL ($\beta = 1.0$)	BEAR ($\beta = 0.6$)
LOW VOL ($\alpha = 1.2$)	$1.2 \times 1.0 = 1.20$	$1.2 \times 0.6 = 0.72$
NORMAL ($\alpha = 1.0$)	$1.0 \times 1.0 = 1.00$	$1.0 \times 0.6 = 0.60$
HIGH VOL ($\alpha = 0.5$)	$0.5 \times 1.0 = 0.50$	$0.5 \times 0.6 = 0.30$

Best conditions (Low Vol + Bull): 1.20× base exposure
Worst conditions (High Vol + Bear): 0.30× base exposure — near-flat book.

4.4 Regime Filter Implementation

```
# --- Volatility Regime ---
spy['vol_20'] = spy['log_return'].rolling(20).std() * np.sqrt(252)
spy['vol_q75'] = spy['vol_20'].rolling(252).quantile(0.75)
spy['vol_q25'] = spy['vol_20'].rolling(252).quantile(0.25)

spy['vol_regime'] = np.where(
    spy['vol_20'] >= spy['vol_q75'], 'HIGH',
    np.where(spy['vol_20'] <= spy['vol_q25'], 'LOW', 'NORMAL')
)

m_vol = {'HIGH': 0.5, 'NORMAL': 1.0, 'LOW': 1.2}

# --- Trend Regime ---
spy['ma50'] = spy['Close'].rolling(50).mean()
spy['trend_regime'] = np.where(spy['Close'].shift(1) > spy['ma50'], 'BULL', 'BEAR')
m_trend = {'BULL': 1.0, 'BEAR': 0.6}
```

```
# --- Combined Multiplier ---
spy['regime_multiplier'] = spy['m_vol'] * spy['m_trend']
```

5 Full Strategy Pipeline Summary

Raw OHLCV Data (50 stocks + SPY, daily)

$\downarrow$

Feature Engineering $\rightarrow$ MA20, vol_20, z_score, 12-1 momentum, log_return

$\downarrow$

Signal 1: Mean Reversion $\rightarrow S_{mr} = +1/0/-1$

Signal 2: Momentum $\rightarrow S_{mom} = +1/0/-1$

$\downarrow$

Hybrid Signal: $S_{hybrid} = 0.5 \times S_{mr} + 0.5 \times S_{mom}$

$\downarrow$

Cross-Sectional Ranking $\rightarrow$ Long top 20%, Short bottom 20%

$\downarrow$

Regime Filter (vol quantile $\times$ SPY/MA50) $\rightarrow m \in [0.30, 1.20]$

$\downarrow$

Final Position: $w_i = \text{signal}_i \times m/n_{\text{active positions}}$

$\downarrow$

Weekly Rebalance $\rightarrow$ P&L after costs (0.10% commission + 0.02% slippage)

6 Assumptions & Findings

- Mean reversion: Z-score extreme deviations ($|z| > 2$) revert within 3–7 trading days on average.
- Momentum persistence: 12-1 momentum exhibits persistence over medium-term horizons (weeks to months).
- Signal complementarity: The two signals are largely uncorrelated. When they agree, conviction and expected alpha are higher.
- Short-term reversal: Excluding the most recent month from momentum avoids known short-term reversal effects.
- High volatility: High-vol environments increase noise and reduce signal reliability for both strategies.
- Market trend: Cross-sectional dispersion and strategy performance are influenced by overall market direction.
- Equal weighting: Avoids concentration risk; each signal contributes equally to the hybrid.